

"Morris" 5-DOF Robotic Arm

Benjamin McEwan

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1 Introduction

Morris

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2 Bill of Materials (BOM)

Table 1: Purchased Parts

Description	Qty
MG996R Servo	5
ESP32 Devkit Type-C	1
PCA9685 16-Channel 12-Bit PWM Driver	1
DC5525 to Screw Terminal Connector DC Power Jack (Female)	1
10A Adjustable DC-DC Step Down Converter	1

3 Analysis

3.1 Forward Kinematics

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3.2 Inverse Kinematics

When calculating the inverse kinematics (IK) of the Morris robot we will consider the robot's spatial frame $\{s\}$ to be placed at the center of the base of the robot and aligned such that the $-y$ direction faces forward, the $-x$ direction faces right, and the $+z$ direction faces upwards.

The n links of the robot are taken as L_n with joints J_n . As link 2 has an L-shaped structure it is more specifically denoted by the horizontal and vertical offsets between the joints J_2 and J_3 , notated as L_{2x} and L_{2y}

The following covers the IK required to position the robot's end-effector at a point $\vec{p} = (p_x, p_y, p_z)$ in 3D space and an orientation ϕ in the 2D space of the robot's wrist. The robot is kinematically decoupled into three independent systems. The base pivot J_1 , the main arm J_2 and J_3 , and the wrist J_4 .

J_1 is considered a 1R mechanism that acts in the x-y plane that only needs to point towards the target point $\vec{p}$, as such the IK angle for J_1 is found via

$$\theta_1 = \text{atan2}(p_y, p_x)$$

J_2 , J_3 , and J_4 are all considered coplanar. In order to apply IK to these joints, the point $\vec{p}$ must be converted into a point $\vec{p}' = (p'_x, p'_y)$ this plane. The x coordinate of this point can be found as magnitude of the x-y plane projection of $\vec{p}$; whilst the y coordinate is simply the z coordinate of $\vec{p}$, therefore

$$\begin{aligned} p'_x &= \sqrt{p_x^2 + p_y^2} \\ p'_y &= p_z \end{aligned}$$

J_2 and J_3 form a 2R mechanism that is decoupled from J_4 . The IK equations of this 2R mechanism must therefore be calculated for a point $\vec{P} = (P_x, P_y)$ which is offset from $\vec{p}'$ by the length L_4 at the angle ϕ , the coordinates of this point are found as

$$\begin{aligned} P_x &= p'_x - L_4 \cos(\phi) \\ P_y &= p'_y - L_4 \sin(\phi) - h \end{aligned}$$

where h is the vertical offset of J_1 from the origin of $\{s\}$. Link 2 possesses an L-shaped structure, as such an "offset angle" θ_0 is established, and the length of the effective link L_2 is calculated from its components,

$$\begin{aligned} \theta_0 &= \text{atan2}(L_{2y}, L_{2x}) \\ L_2 &= \sqrt{L_{2x}^2 + L_{2y}^2} \end{aligned}$$

The distance between J_2 and the target position $\vec{P}$ is denoted d and calculated as

$$d = \sqrt{P_x^2 + P_y^2}$$

Using the above variables, θ_2 and θ_3 can then be seen via the law of cosines to be

$$\begin{aligned}\theta_2 &= \theta_0 + \text{atan2}(P_y, P_x) + \arccos\left(\frac{L_2^2 + d^2 - L_3^2}{2L_2d}\right) \\ \theta_3 &= \arccos\left(\frac{L_2^2 + L_3^2 - d^2}{2L_2L_3}\right) - \theta_0 - \frac{\pi}{2}\end{aligned}$$

From which the angle θ_4 is defined as

$$\phi - (\theta_2 + \theta_3) + \frac{\pi}{2}$$

3.2.1 Implementation of Inverse Kinematics

The calculated angles θ_1 , θ_2 , θ_3 , and θ_4 are the angles between each link and as such they cannot be directly inputted into the robot's servo joints. For this task the auxilliary angles α_n are defined as

$$\begin{aligned}\alpha_1 &= \pi + \theta_1 \\ \alpha_2 &= \pi + \theta_2 \\ \alpha_3 &= \theta_3 - \frac{\pi}{2} \\ \alpha_4 &= \frac{\pi}{2} - \theta_4\end{aligned}$$

It is also recommended that a clamping function is introduced into any implementation of these angles to prevent damagining the servo motors.

3.2.2 Workspace Analysis

The arm has a limited range and as such if a target point $\vec{p}$ is outside of this range the θ_n functions may fail to produce values. For a point to be in the workspace of the robot the expression ///